Installing AGAMAb on a Windows machine

Here we explain how to install the main AGAMAb library (agamab.dll) and its Python extension (agamab.pyd) on a Windows computer using the MSVC IDE. The IDE makes updating AGAMAb fast and easy. Before you start, install gsl and pybind11.

After cloning the repository go to the src folder and delete Py_agama.cpp. Then copy in the contents of ..\windows . Now within the VS IDE click to open ‘a project or solution’. Navigate to the src folder and click on agama.sln

Then click on ‘build’ on the ‘build’ drop- down menu. All being well agamab.dll and agamab.lib will appear in the x64\release subfolder of the src folder. If this folder in on your PATH you are ready to go. If not, add either it to PATH or copy agamab.dll to a folder that is on the PATH. agamab.lib is used by the linker. A suitable script vscl.bat for compiling and linking to AGAMAb reads

```
@echo off

cl.exe /c /openmp /Ih:\u\c\agamab\src -EHsc /std:c++20 %1.cpp

@if not %ERRORLEVEL% == 0 (goto end)

link %1.obj %2 %3 %4 %5 agamab.lib /LIBPATH:\u\c\agamab\src\x64\release /NOIMPLIB

del %1.obj

del %1.exp

:end
```

You’ll need to change the h:\u\c to the path to the agamab folder on your machine.

If build doesn’t work first time go to the bottom of the project tab and click on ‘properties’ and review options.

Once agamab.dll is made, close the agamab project and go to the Py subfolder of the agamab folder and there click on Py_agama.sln . Go to the bottom of the ‘project’ drop-down and click on ‘configuration’. Click on ‘general’ and change the ‘Output Directory’ to the agamab subfolder of the folder that PYTHONPATH points to: I’ve assigned this to the root of drive k:. Also in ‘general’ change ‘TargetName’ to agamab. The ‘Configuration Type’ should already be Dynamic Library. At the top of ‘Advanced’ make sure the TargetExtension is .pyd In VC++ Directories check that Include Directories include the py_bind11 and python directories: with k: pointing to the PYTHONPATH folder, pybind11 is at k:\pybind11

In ‘C/C++’ check that Additional Include Directories covers the agamab\src folder where AGAMAb’s header files are located

In ‘Linker’ ensure that Additional Library Directories’ covers the location of the Python library and the location of agamab.lib

All that done, the ‘build’ tab should put agamab.pyd in the agamab subfolder at the end of PYTHONPATH

Finally copy __init__.py from the src\Py folder to the PYTHONPATH agamab subfolder after editing it to ensure that the argument of add_dll_directory() is the location of agamab.dll (For some reason Python doesn’t use the PATH to find this). You should now be able to ‘import agamab as whatever’.